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The sporangia with spores are found in sporophylls. The sporophylls organise to form cones or strobilus. The spores germinates to produce haploid, multicellular heart shaped independent gametophyte called prothallus. Sexual reproduction is Oogamous. The life cycle shows Alternation of generation.

The term stele includes central cylinder of vascular tissues comprising xylem, phloem, pericycle, endodermis and pith . There are two major types of stele namely Protostele and Siphonostele.

Selaginella belongs to the class Lycopsida. The plant body is sporophyte. It is differentiated into stem, leaf, rhizophore and roots. Heterospory is found and two types of spores namely microspores and megaspores are produced in sporangia. The microsporangia and megasporania are borne on sporophylls. The sporophylls are organized to form cone. Sexual reproduction is oogamous. Alternation of generation is present.

Adiantum belongs to Pteropsida. The sporophyte is differentiated into root, rhizome and leaves. The spores are produced in sporangia and is covered by false indusium. The sexual reproduction is oogamous and sex organs (antheridum and archegonium) are produced on prothallus. Alternation of generation is present.

Gymnosperms are naked seed producing plants. The plant body is sporophyte and it is the dominant phase. Coralloid roots are found in *Cycas*. The roots of *Pinus* possess Mycorrhizal association .Two types of branches called Long shoot and dwarf shoot are present. Stem shows secondary growth. Spores are produced in cones. Pollen tube helps in fertilization.. The endosperm is haploid . Alternation of generation is present

Cycas belongs to Cycadopsida. The plant body is sporophyte and looks like a small palm tree. Apart from Taproot Coralloid roots are present. It is dioecious, Microsporophylls are organized into male cone. Ovules are borne on megasporophylls which are not organized into cone. Fertilization results in zygote and it develops into embryo. Alternation of generation is present.

Pinus belongs to Coniferopsida.. The plant body is sporophyte and is differentiated into root, stem and leaves. The main stem is branched. The branches are dimorphic with long and short branches. It is monoecious, heterosporous and produces two types of spores called microspores and megaspores. Alternation of generation is present.

Angiosperms are highly evolved plant group and their ovules remain enclosed in an ovary. A wide range of habit is present.. These include trees, shrubs, herbs, climbers, lianas. Double fertilization is present. The endosperm is triploid. They are classified into Dicotyledons and Monocotyledons.

Evaluation

1. Which of the plant group has gametophyte as a dominant phase?
   a. Pteridophytes
   b. Bryophytes
   c. Gymnosperm
   d. Angiosperm
2. Which of following represent gametophytic generation in pteridophytes?
   a. Prothallus
   b. Thallus